Short Analysis Pipeline

BIOLM0051

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Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

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Introduction

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Methods

The FASTQ files in this report are small and the use of HPC resources in this case is not necessary. However, this workflow is designed to scale easily to similar analyses with much larger datasets. All steps here are able to run on an HPC cluster without modification, demonstrating reproduciility, automation, and capacity for high-throughput analyses.

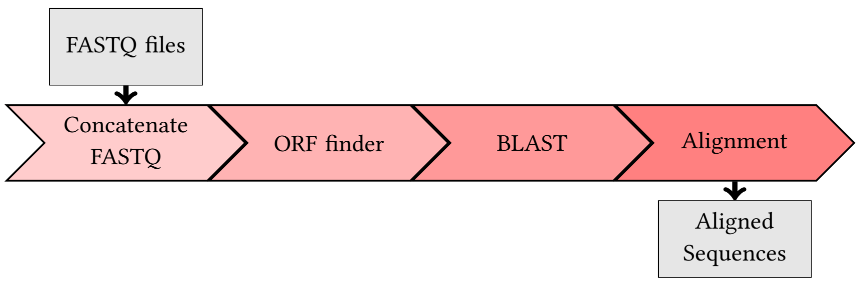

Figure 1: Workflow of Snakemake analysis pipeline.

Some python code:

```python
from Bio import SeqIO
for record in SeqIO.parse("sequence.fasta", "fasta"):
    print(record.id, record.seq.translate())
```

Some Bash script:

```bash
#!/bin/bash
#SBATCH --job-name=blast_job
#SBATCH --partition=teach_cpu
#SBATCH --time=01:00:00
#SBATCH --mem=2000M

blastn -db nt -query sequence.fa -out results.out -outfmt 7 -remote
```

Results

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Table 1: Top 15 BLASTP hits for sample OMRGCv2f_11922010_1 against SwissProt.

Query	Subject	% Identity	Alignment Length	E-value	Bitscore
OMRGCv2f_11922010_1	A5UYW4.1	55.56	63	1.3e-17	85.1
OMRGCv2f_11922010_1	A7NS65.1	55.56	63	1.8e-17	83.5

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```python
import pandas as pd
data = pd.read_csv("expression.csv")
normalized = data.apply(lambda x: np.log2(x+1))
```

Alignment results using Clustal Omega on some test data:

```
test1      .MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANK    54
test2      .MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDANK    54
test3      KAGSAKKGATLFKTRCLQCHTVEKGGPHKVGPNLHGIFGRHSGQAEGYSYTDANI    59
consensus  .MGDVEKGKKIF..KC.QCHTVEKGGKHKTGPNLHGLFGRKTGQA.GYSYTDANK


test1      NKGIIWGEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE   105
test2      NKGITWGEDTLMEYLENPKKYIPGTKMIFAGIKKKGERADLIAYLKKATNE   105
test3      KKNVLWDENNMSEYLTNPKKYIPGTKMAFGGLKKEKDRNDLITYLK.....   105
consensus  NKGI.WGEDTLMEYLENPKKYIPGTKMIF.GIKKK.ERADLIAYLKKATNE

           X  non-conserved
           X  ≥ 50% conserved
           X  ≥ 80% conserved
```

Figure 2: Alignment using Clustal Omega and Identity mode in texshade

Discussion

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui.

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References

[1] WJ Kent. Blat—the blast-like alignment tool. *Genome research*, 12(4):656–664, 2002.

[2] MJ Wolin, TL Miller, and CS Stewart. Microbe-microbe interactions. In *The rumen microbial ecosystem*, pages 467–491. Springer, 1997.

[3] B Baker, P Zambryski, B Staskawicz, and SP Dinesh-Kumar. Signaling in plant-microbe interactions. *Science*, 276(5313):726–733, 1997.

Acknowledgments

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Supplementary Data

Supplementary data can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git